SIMATS ENGINEERING

ASSESSMENT TOOL 2

COURSE NAME: Cryptography and Network Security COURSE CODE: CSA5116

COURSE OUTCOME COVERED

CO1: Apply classical encryption techniques using symmetric and asymmetric ciphers to encrypt and decrypt data for the ensuring data security. (BL3, BL4)

Assessment Tool Weightage: 20%

QUESTIONS: 5 **Total Marks:50**

Annexure A

CONCEPT MAPPING

Code of Conduct:

I Ganesh.B / 192511066 *(Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.*
I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student: B. Ganesh

CSA5116 - Cryptography and Network security

CO1 - AT2 - Concept Map

1) Security Attacks, Threats, Risks and security Goals :

Concept Map :

security Attack -> Threat -> Risk -> Security Goals

Security goals -> Confidentiality + Integrity + Availability.

1) Security Attack :

It is an attempt to access, Modify or destroy or Misuse Information.

Ex : phishing, Malware, and Dos attacks.

2) Threat :

A Threat is a possible danger that can exploit a weakness in a System.

Examples : phishing creates a Threat of password Theft.

3) Risk :

Risk is the possibility of loss or damage when a Threat exploits a vulnerability.

Ex : password stolen -> unAuthorized access -> Account / data Loss.

4) H) Confidentiality :-

- prevents unauthorized access to Information
- Ensure data is available only to authorized users.
- Ex :- Encryption protects banking Info

5) Availability :-

- Ensures Systems and data are available when required.
- Ex -> Backup Systems can help Maintain services during an attack.

Attack -> Risk Mapping :-

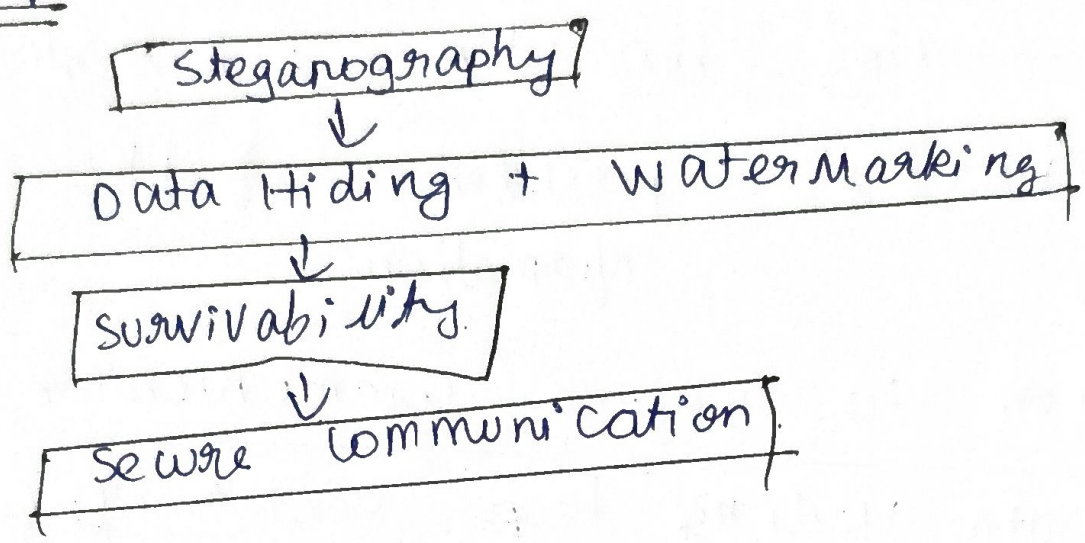
Attack	Threat / Risk	security goal
phishing	Password theft	Confidentiality
Malware	Data Modification	Integrity
DOS	service disruption	Availability.

Conclusion :-

Security attacks create threats, which Can lead to risks. the CIA triad - confidentiality Integrity, and Availability - helps reduce these risks and protect Information.

4) Steganography :- Data Hiding, watermarking and survivability :-

Concept Map



1) Data Hiding :-

It means concealing secret information inside another Medium such as an image, audio or video.

purpose :-

-> Hides the existence of secret Information
-> supports Confidential communication.

2) Watermarking :-

It embeds Information into digital content such as images, audio or video.

purpose :-

-> protects ownership
-> Helps Identify the creator
-> supports copyright protection.

3) survivability :-

It means that the hidden or embedded Information remains available after modifications.

Relationship Between the Concepts :-

- Data Hiding -> secrecy
- Watermarking -> ownership and Authenticity
- Survivability -> persistence of Hidden Information.

Contribution to Secure communication :-

1. Data Hiding keeps secret Information Concealed.
2. Survivability ensures the embedded Info remains effective after Modifications.
3. watermarking protects digital ownership
4. Together they improve protection of digital Information.

Conclusion :-

Steganography provides secure Information hiding.